Report:

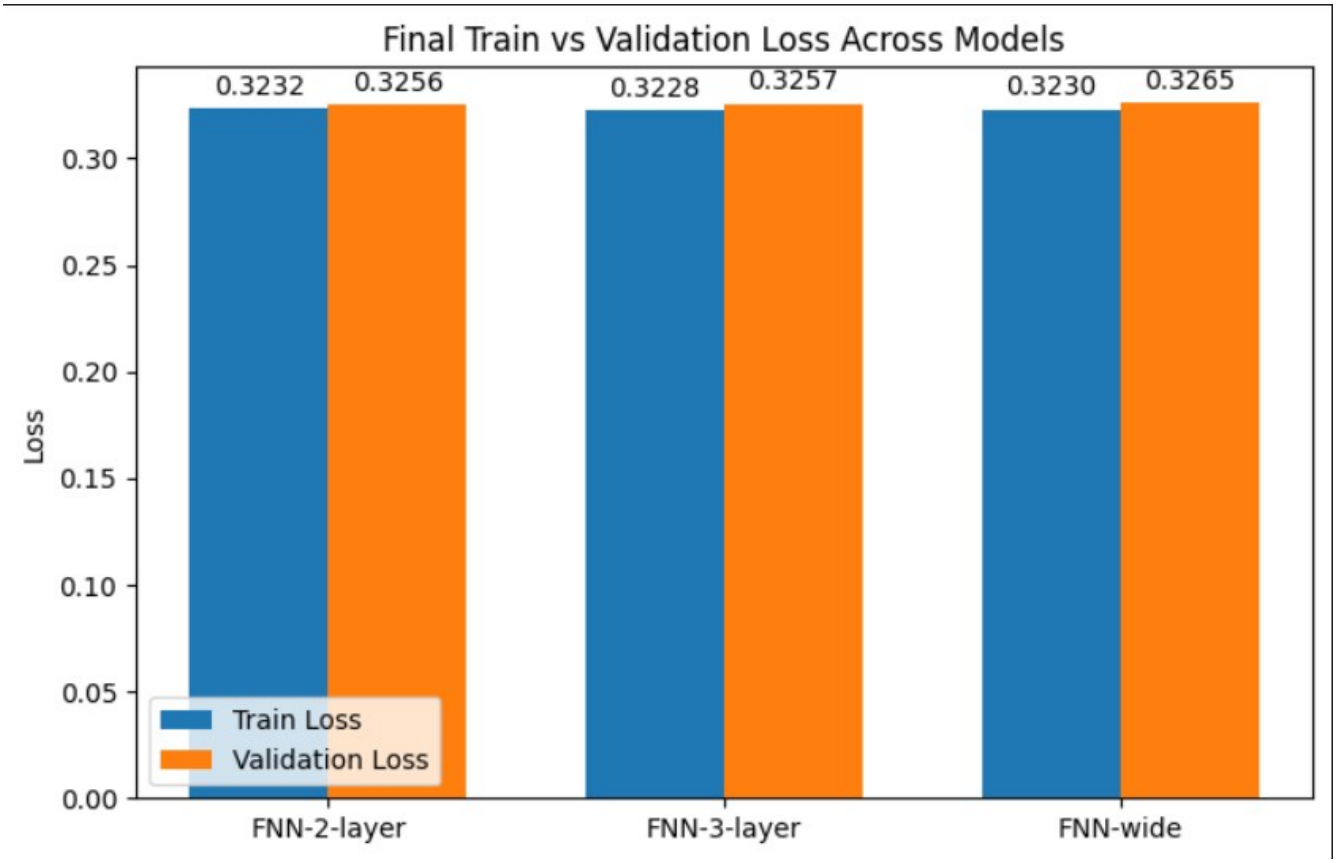
There are 4 Neural Network Architecture implemented in this assignment.
Here Batch Size: 64
Optimizer: Adam
Number of epochs: 20
Learning Rate: 0.001

1st Neural Network Architecture:
The number of hidden layers → 2
The number of neurons →
128 in 1st hidden layer, 64 in 2nd hidden layer
loss function → Binary Cross Entropy Loss
activation function→
ReLU (Rectified Linear Unit) for hidden layer to introduce non-linearity and improve learning efficiency.

2nd Neural Network Architecture:
The number of hidden layers → 3
The number of neurons →
256 in 1st hidden layer, 128 in 2nd hidden layer, 64 in 3rd hidden layer
loss function → Binary Cross Entropy Loss
activation function→
ReLU (Rectified Linear Unit) for hidden layer to introduce non-linearity and improve learning efficiency.

3rd Neural Network Architecture:
The number of hidden layers → 2
The number of neurons →
200 in 1st hidden layer, 100 in 2nd hidden layer
loss function → Binary Cross Entropy Loss
activation function→
ReLU (Rectified Linear Unit) for hidden layer to introduce non-linearity and improve learning efficiency.

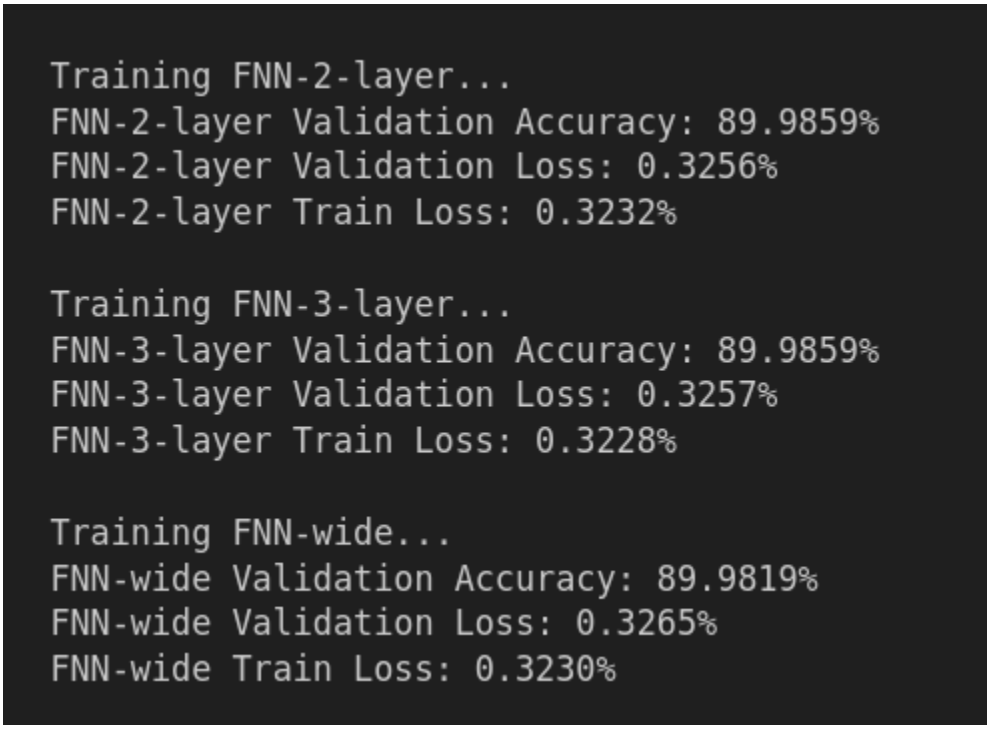
2. Training and Validation loss Plot:



3. Final Selected Architecture:

Selected Model: FNN-2-Layer

Through the validation loss for all the models are almost equal, FNN-2-Layer architecture is selected as the validation loss of this model is least.

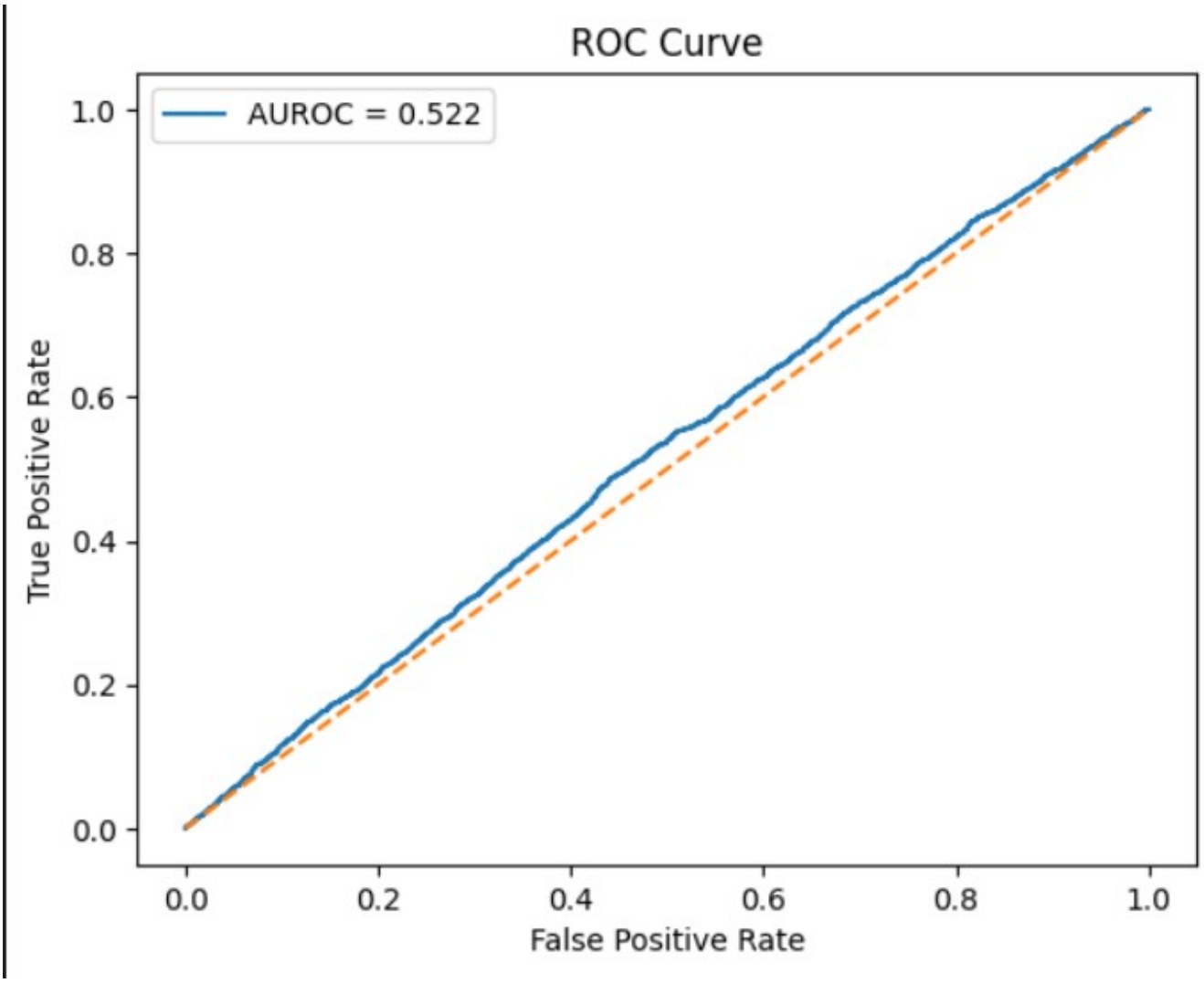


The reason of choosing Validation Loss as indicator:
Validation loss measures how well the model performs on new or unseen data. Low validation loss means that the model does not overfit the data. It balances accuracy and generalization. That's why I chose the model with the least validation loss.

4. The Reason Behind Unexpected Output:

Accuracy : 0.8999
Precision: 0.4499
F1-score: 0.4736
AUROC: 0.5159

AUROC is almost 52%, which is really near to random guessing. Here is the graph.



The reason is that the dataset is imbalanced. This is the number of 1 and 0 in the target column–
Number of 1 in target column: 16985
Number of 0 in target column: 152640

The positive and negative ratio in the target column is 1:9. This creates a bias and the outputs are affected. That's why no model gives good results even after data processing.